Small-Data Surrogate Shock-Aware Designed (S3) models: Gradient-Free Residual Adaptation with Adaptive Conformal Uncertainty for Engineering Time Series

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Abstract

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1. Introduction

1.1. Key Contributions

1. We formulate small-data engineering forecasting as a *surrogate residual adaptation* problem and propose a gradient-free correction architecture that freezes the baseline forecaster and learns only a low-capacity residual readout, thereby prioritizing variance control and convex optimization stability over end-to-end parameter proliferation [7, 13].
2. We introduce two complementary model realizations—*S3-Forecaster*, based on fixed-reservoir Echo State dynamics under spectral radius constraints, and *S3-FastSketch*, based on deterministic causal multiscale sketches—to separate temporal representation from lightweight residual decoding in genuinely short-series regimes [41, 42, 44].
3. We develop a shock-aware residual control mechanism based on out-of-bag gating and bounded shrinkage, motivated by the documented adaptation lag of classical online updaters under concept drift and by the need to suppress variance amplification when apparent residual structure is not stable enough to be learnable [25, 26].
4. We integrate a sequential Adaptive Conformal Inference layer for uncertainty quantification, so that prediction intervals are calibrated by recent nonconformity scores and retain distribution-free predictive validity despite exchangeability violation in dependent engineering sequences [29, 36, 37].
5. We adopt a strict chronological train–calibration–test protocol designed to preserve temporal order and avoid information leakage and look-ahead bias, aligning evaluation with the actual decision task faced in forecast deployment rather than with i.i.d. cross-validation surrogates [39, 40].

2. Materials and Methods

2.1. Problem Formulation

Let

$$\boldsymbol{y}_{1:N} = (y_1, \ldots, y_N)^\top \in \mathbb{R}^N \tag{1}$$

denote a univariate engineering time series with monthly sampling frequency and limited historical availability, $N < 200$. The forecasting problem is defined with respect to the natural filtration

$$\mathcal{F}_t = \sigma(y_1, \ldots, y_t), \tag{2}$$

which represents all target information observed up to time t. A valid one-step-ahead predictor must satisfy

$$\widehat{y}_{t|t-1} \in \mathcal{F}_{t-1}, \tag{3}$$

such that y_t cannot influence its own forecast. The proposed family follows a residual-adaptation decomposition:

$$\widehat{y}_{t|t-1} = \widehat{y}^{\text{base}}_{t|t-1} + \widehat{\Delta}_{t|t-1}, \tag{4}$$

where $\widehat{y}^{\text{base}}_{t|t-1}$ is generated by a frozen causal prior and $\widehat{\Delta}_{t|t-1}$ is a data-efficient residual correction. The causal base residual is

$$e_t = y_t - \widehat{y}^{\text{base}}_{t|t-1}. \tag{5}$$

The proposed models differ only in the representation used to estimate the predictable component of e_t:

$$\text{S3-Forecaster:} \quad e_{1:t-1} \longmapsto \boldsymbol{h}_{t-1} \longmapsto \widehat{r}_{t|t-1}, \tag{6}$$

$$\text{S3-FastSketch:} \quad e_{1:t-1} \longmapsto \boldsymbol{\phi}_{t-1} \longmapsto \widehat{r}_{t|t-1}. \tag{7}$$

Neither architecture backpropagates through the causal prior or through its nonlinear feature extractor. The learned component is restricted to a regularized linear readout and a low-dimensional calibration mechanism.

2.2. Frozen Causal Forecast Prior

Let

$$f_{\text{base}} : \bigcup_{t \geq 1} \mathbb{R}^t \longrightarrow \mathbb{R} \tag{8}$$

denote a statistical, foundation-model, or heuristic forecasting operator whose parameters are frozen during residual adaptation. Its strictly causal output is

$$\widehat{y}^{\text{base}}_{t|t-1} = f_{\text{base}}\left(y_1, \ldots, y_{t-1}; \boldsymbol{\theta}_{\text{base}}\right). \tag{9}$$

The generic formulation permits the use of rolling estimators, ETS, Prophet, ARIMA, or a zero-shot time-series foundation model. The residual adapter does not alter $\boldsymbol{\theta}_{\text{base}}$. Consequently, adaptation cost is independent of the number of parameters in the frozen prior. For the lightweight default prior, a causal rolling level estimate is defined as

$$\widehat{\mu}_{t-1}^{(w)} = \frac{1}{|\mathcal{W}_{t-1}|} \sum_{j \in \mathcal{W}_{t-1}} y_j, \qquad \mathcal{W}_{t-1} = \{\max(1, t-w), \dots, t-1\}. \tag{10}$$

When sufficient monthly history is available, a causal seasonal correction may be added:

$$\widehat{s}_{m(t)} = \frac{1}{|\mathcal{I}_{m(t)}|} \sum_{j \in \mathcal{I}_{m(t)}} \left(y_j - \widehat{\mu}_{j-1}^{(w)} \right), \tag{11}$$

where $\mathcal{I}_{m(t)}$ contains only pre-t observations belonging to the same calendar month as t. The resulting prior is

$$\widehat{y}_{t|t-1}^{\text{base}} = \widehat{\mu}_{t-1}^{(w)} + \widehat{s}_{m(t)}. \tag{12}$$

2.3. S3-Forecaster: Fixed Reservoir Residual Adaptation

2.3.1. Residual Normalization

The residual stream is standardized using statistics estimated exclusively from the training block:

$$\widetilde{e}_t = \frac{e_t - \widehat{m}_e}{\widehat{q}_{0.75}(e) - \widehat{q}_{0.25}(e) + \varepsilon_s}, \qquad t \in \mathcal{D}_{\text{tr}}, \tag{13}$$

where $\widehat{m}_e$ is the training median and $\varepsilon_s > 0$ prevents division by zero. The same frozen transformation is used for calibration and testing.

2.3.2. Echo State Representation

Let R denote the reservoir dimension. The recurrent state is

$$\boldsymbol{h}_t \in [-1, 1]^R. \tag{14}$$

A leaky ESN update is defined by

$$\widetilde{\boldsymbol{h}}_t = \tanh\left(\boldsymbol{W}_{\text{in}} \begin{bmatrix} 1 \\ \widetilde{e}_t \end{bmatrix} + \boldsymbol{W}_{\text{res}} \boldsymbol{h}_{t-1} \right), \tag{15}$$

$$\boldsymbol{h}_t = (1 - \lambda_{\text{leak}}) \boldsymbol{h}_{t-1} + \lambda_{\text{leak}} \widetilde{\boldsymbol{h}}_t, \qquad \lambda_{\text{leak}} \in (0, 1]. \tag{16}$$

The fixed matrices satisfy

$$\boldsymbol{W}_{\text{in}} \in \mathbb{R}^{R \times 2}, \tag{17}$$

$$\boldsymbol{W}_{\text{res}} \in \mathbb{R}^{R \times R}. \tag{18}$$

They are sampled once from seed-controlled random distributions and are not updated during training. Let $\widetilde{\boldsymbol{W}}_{\text{res}}$ be an initially sampled recurrent matrix. Spectral normalization is performed as

$$\boldsymbol{W}_{\text{res}} = \rho_\star \frac{\widetilde{\boldsymbol{W}}_{\text{res}}}{\rho\left(\widetilde{\boldsymbol{W}}_{\text{res}}\right) + \varepsilon_\rho}, \qquad 0 < \rho_\star < 1.5, \tag{19}$$

where $\rho(\cdot)$ denotes spectral radius. The stronger contraction condition

$$\|\boldsymbol{W}_{\text{res}}\|_2 < 1.5 \tag{20}$$

is sufficient for asymptotic forgetting of initial states under a 1-Lipschitz activation. In practice, $\rho(\boldsymbol{W}_{\text{res}}) < 1.5$ is used as the standard reservoir-design heuristic. Its purpose is to prevent uncontrolled recurrent amplification while retaining a fading memory of the residual history. To supplement the nonlinear reservoir with explicit short-term information, an autoregressive residual vector is constructed:

$$\boldsymbol{\ell}_t^{(p)} = [e_t,\, e_{t-1},\, \dots,\, e_{t-p+1}]^\top \in \mathbb{R}^p. \tag{21}$$

The complete S3 representation is

$$\boldsymbol{z}_t = \left[1,\, \boldsymbol{h}_t^\top,\, \left(\boldsymbol{\ell}_t^{(p)}\right)^\top\right]^\top \in \mathbb{R}^P, \qquad P = 1 + R + p. \tag{22}$$

2.3.3. Bayesian Ridge Residual Readout

For one-step-ahead forecasting, the readout assumes

$$e_{t+1} = \boldsymbol{w}^\top \boldsymbol{z}_t + \xi_{t+1}, \qquad \xi_{t+1} \sim \mathcal{N}(0, \beta^{-1}). \tag{23}$$

A zero-mean isotropic Gaussian prior is imposed:

$$p(\boldsymbol{w} \mid \alpha) = \mathcal{N}\left(\boldsymbol{0}, \alpha^{-1}\boldsymbol{I}_P\right). \tag{24}$$

Let

$$\boldsymbol{Z} = [\boldsymbol{z}_{t_1},\, \dots,\, \boldsymbol{z}_{t_n}]^\top \in \mathbb{R}^{n\times P}, \tag{25}$$

$$\boldsymbol{r} = [e_{t_1+1},\, \dots,\, e_{t_n+1}]^\top \in \mathbb{R}^n. \tag{26}$$

The negative log-posterior, excluding constants independent of $\boldsymbol{w}$, is

$$\mathcal{J}(\boldsymbol{w}) = \frac{\beta}{2}\|\boldsymbol{r} - \boldsymbol{Z}\boldsymbol{w}\|_2^2 + \frac{\alpha}{2}\|\boldsymbol{w}\|_2^2. \tag{27}$$

The Hessian is

$$\nabla_{\boldsymbol{w}}^2 \mathcal{J}(\boldsymbol{w}) = \beta\boldsymbol{Z}^\top\boldsymbol{Z} + \alpha\boldsymbol{I}_P \succ 0 \tag{28}$$

for $\alpha > 0$. Therefore, the objective is strictly convex and has a unique global minimizer. Setting the gradient to zero gives

$$\nabla_{\boldsymbol{w}} \mathcal{J}(\boldsymbol{w}) = -\beta\boldsymbol{Z}^\top(\boldsymbol{r} - \boldsymbol{Z}\boldsymbol{w}) + \alpha\boldsymbol{w} = \boldsymbol{0}, \tag{29}$$

$$\left(\beta\boldsymbol{Z}^\top\boldsymbol{Z} + \alpha\boldsymbol{I}_P\right)\boldsymbol{w} = \beta\boldsymbol{Z}^\top\boldsymbol{r}. \tag{30}$$

Hence,

$$\widehat{\boldsymbol{w}}_{\text{MAP}} = \left(\beta\boldsymbol{Z}^\top\boldsymbol{Z} + \alpha\boldsymbol{I}_P\right)^{-1}\beta\boldsymbol{Z}^\top\boldsymbol{r}. \tag{31}$$

Equivalently, defining $\lambda_R = \alpha/\beta$,

$$\widehat{\boldsymbol{w}}_{\mathrm{MAP}} = \left(\boldsymbol{Z}^\top \boldsymbol{Z} + \lambda_R \boldsymbol{I}_P\right)^{-1} \boldsymbol{Z}^\top \boldsymbol{r}. \tag{32}$$

The complete posterior is

$$p(\boldsymbol{w} \mid \boldsymbol{r}, \boldsymbol{Z}, \alpha, \beta) = \mathcal{N}\left(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w\right), \tag{33}$$

$$\boldsymbol{\Sigma}_w = \left(\alpha \boldsymbol{I}_P + \beta \boldsymbol{Z}^\top \boldsymbol{Z}\right)^{-1}, \tag{34}$$

$$\boldsymbol{\mu}_w = \beta \boldsymbol{\Sigma}_w \boldsymbol{Z}^\top \boldsymbol{r}. \tag{35}$$

The one-step residual correction is

$$\widehat{r}_{t+1|t} = \boldsymbol{\mu}_w^\top \boldsymbol{z}_t. \tag{36}$$

The Bayesian derivation provides a probabilistic interpretation of the regularized readout. When fixed precision parameters are used, its posterior mode is numerically equivalent to conventional Ridge regression. Lasso and Elastic Net may be considered as convex sensitivity variants, but they must not be described as Bayesian Ridge unless the corresponding prior interpretation is explicitly introduced.

2.3.4. Volatility-Sensitive OOB Gate

The residual correction is not assumed to be uniformly reliable. A causal volatility process is therefore computed from the prior residuals. Let

$$\mu_t^{(e)} = (1-\lambda_v)\mu_{t-1}^{(e)} + \lambda_v e_t, \tag{37}$$

$$q_t^{(e)} = (1-\lambda_v)q_{t-1}^{(e)} + \lambda_v \left(e_t - \mu_{t-1}^{(e)}\right)^2, \tag{38}$$

$$v_t = \sqrt{q_t^{(e)} + \varepsilon_v}. \tag{39}$$

The volatility is robustly standardized using training-only statistics:

$$\widetilde{v}_t = \frac{v_t - \mathrm{median}_{j \in \mathcal{D}_{\mathrm{tr}}}(v_j)}{\mathrm{IQR}_{j \in \mathcal{D}_{\mathrm{tr}}}(v_j)/1.349 + \varepsilon_v}. \tag{40}$$

The gate is

$$g_t(a,b) = \sigma\left(a(\widetilde{v}_t - b)\right) = \frac{1}{1+\exp[-a(\widetilde{v}_t - b)]}, \qquad g_t \in (0,1). \tag{41}$$

The ESN and its readout are fitted on $\mathcal{D}_{\mathrm{tr}}$. Keeping these components fixed, the two gate parameters are estimated exclusively on the chronological calibration block:

$$(\widehat{a}, \widehat{b}) = \underset{a \in [a_{\min}, a_{\max}],\, b \in [b_{\min}, b_{\max}]}{\arg\min} \frac{1}{|\mathcal{D}_{\mathrm{cal}}|} \sum_{t \in \mathcal{D}_{\mathrm{cal}}} \left| e_t - g_{t-1}(a,b)\widehat{r}_{t|t-1} \right|. \tag{42}$$

This is an out-of-bag optimization with respect to the readout because no calibration observation is used to estimate $\boldsymbol{w}$. The gate is a two-parameter calibration mechanism rather than a second high-capacity predictor. The final S3 point forecast is

$$\widehat{y}_{t|t-1}^{\mathrm{S3}} = \widehat{y}_{t|t-1}^{\mathrm{base}} + g_{t-1}(\widehat{a}, \widehat{b})\widehat{r}_{t|t-1}. \tag{43}$$

The term *gradient-free* refers to the absence of end-to-end backpropagation through the causal prior and reservoir. The bounded low-dimensional calibration of (a, b) is not conflated with neural gradient-based fine-tuning.

2.4. S3-FastSketch: Lightweight Multiscale Residual Adaptation

2.4.1. Causal Multiscale Feature Map

S3-FastSketch replaces the recurrent reservoir by a deterministic causal map

$$\Phi : e_{1:t} \longmapsto \boldsymbol{\phi}_t \in \mathbb{R}^{D_\phi}. \tag{44}$$

The representation is

$$\boldsymbol{\phi}_t = \left[\boldsymbol{\ell}_t^\top, \boldsymbol{a}_t^\top, \boldsymbol{v}_t^\top, \boldsymbol{d}_t^\top, \boldsymbol{c}_t^\top, \boldsymbol{s}_t^\top\right]^\top. \tag{45}$$

Let $\mathcal{Q} = \{q_1, \ldots, q_Q\}$ be a set of causal smoothing spans and $\mathcal{K} = \{k_1, \ldots, k_K\}$ a set of local filter scales.

Residual lag block.. For p autoregressive lags,

$$\boldsymbol{\ell}_t = [e_{t-1}, e_{t-2}, \ldots, e_{t-p}]^\top \in \mathbb{R}^p. \tag{46}$$

Exponentially weighted residual block.. For each $q \in \mathcal{Q}$,

$$a_t^{(q)} = (1 - \eta_q) a_{t-1}^{(q)} + \eta_q e_t, \tag{47}$$

$$\eta_q = \frac{2}{q+1}. \tag{48}$$

The block is

$$\boldsymbol{a}_t = \left[a_t^{(q_1)}, \ldots, a_t^{(q_Q)}\right]^\top. \tag{49}$$

Residual-magnitude block.. The multiscale magnitude summaries are

$$v_t^{(q)} = (1 - \eta_q) v_{t-1}^{(q)} + \eta_q |e_t|, \tag{50}$$

and

$$\boldsymbol{v}_t = \left[v_t^{(q_1)}, \ldots, v_t^{(q_Q)}\right]^\top. \tag{51}$$

Local-difference block.. For $q \in \mathcal{Q}$,

$$d_t^{(q)} = \begin{cases} e_t - e_{t-q}, & t > q, \\ 0, & t \le q. \end{cases} \tag{52}$$

Thus,

$$\boldsymbol{d}_t = \left[d_t^{(q_1)}, \ldots, d_t^{(q_Q)}\right]^\top. \tag{53}$$

Fixed causal convolutional block.. For each $k \in \mathcal{K}$, define the causal window

$$\mathcal{W}_{t,k} = \{\max(1, t-k+1), \dots, t\}. \tag{54}$$

Its local mean is

$$c_{t,k}^{(\text{mean})} = \frac{1}{|\mathcal{W}_{t,k}|} \sum_{j \in \mathcal{W}_{t,k}} e_j. \tag{55}$$

Its endpoint slope is

$$c_{t,k}^{(\text{slope})} = e_t - e_{\max(1,t-k+1)}. \tag{56}$$

Let $\mathcal{W}_{t,k}^{L}$ and $\mathcal{W}_{t,k}^{R}$ denote the first and second halves of $\mathcal{W}_{t,k}$. The local contrast is

$$c_{t,k}^{(\text{contrast})} = \frac{1}{|\mathcal{W}_{t,k}^{R}|} \sum_{j \in \mathcal{W}_{t,k}^{R}} e_j - \frac{1}{|\mathcal{W}_{t,k}^{L}|} \sum_{j \in \mathcal{W}_{t,k}^{L}} e_j. \tag{57}$$

The local absolute energy is

$$c_{t,k}^{(\text{energy})} = \frac{1}{|\mathcal{W}_{t,k}|} \sum_{j \in \mathcal{W}_{t,k}} |e_j|. \tag{58}$$

The convolutional block is the concatenation

$$\boldsymbol{c}_t = \operatorname{concat}_{k \in \mathcal{K}} \left[c_{t,k}^{(\text{mean})}, c_{t,k}^{(\text{slope})}, c_{t,k}^{(\text{contrast})}, c_{t,k}^{(\text{energy})} \right]. \tag{59}$$

Seasonal and calendar block.. For monthly seasonality $m = 12$,

$$\boldsymbol{s}_t = \left[e_{t-m},\, e_t - e_{t-m},\, \sin\left(\frac{2\pi t}{m}\right),\, \cos\left(\frac{2\pi t}{m}\right) \right]^{\top}, \tag{60}$$

where unavailable seasonal lags are initialized to zero. Each feature dimension is standardized using training medians and interquartile ranges:

$$\widetilde{\phi}_{t,j} = \frac{\phi_{t,j} - \operatorname{median}_{u \in \mathcal{D}_{\text{tr}}}(\phi_{u,j})}{\operatorname{IQR}_{u \in \mathcal{D}_{\text{tr}}}(\phi_{u,j})/1.349 + \varepsilon_{\phi}}. \tag{61}$$

2.4.2. Regularized Residual Readout

For a forecast horizon H, define

$$\boldsymbol{r}_{t+1:t+H} = [e_{t+1}, \dots, e_{t+H}]^{\top} \in \mathbb{R}^{H}. \tag{62}$$

The multivariate residual readout is

$$\widehat{\boldsymbol{r}}_{t+1:t+H|t} = \boldsymbol{W}^{\top} \widetilde{\boldsymbol{\phi}}_t, \qquad \boldsymbol{W} \in \mathbb{R}^{D_{\phi} \times H}. \tag{63}$$

Let

$$\boldsymbol{\Phi} \in \mathbb{R}^{n_{\text{tr}} \times D_{\phi}}, \tag{64}$$

$$\boldsymbol{R} \in \mathbb{R}^{n_{\text{tr}} \times H} \tag{65}$$

contain the training sketches and future residual targets. The readout solves

$$\widehat{\boldsymbol{W}} = \arg\min_{\boldsymbol{W}} \|\boldsymbol{R} - \boldsymbol{\Phi}\boldsymbol{W}\|_F^2 + \lambda_F \|\boldsymbol{W}\|_F^2. \tag{66}$$

Its closed-form solution is

$$\widehat{\boldsymbol{W}} = \left(\boldsymbol{\Phi}^{\top}\boldsymbol{\Phi} + \lambda_F \boldsymbol{I}_{D_{\phi}} \right)^{-1} \boldsymbol{\Phi}^{\top} \boldsymbol{R}. \tag{67}$$

2.4.3. Projected Calibration Shrinkage

The raw residual forecast may be unstable when the calibration block is small. For each horizon h, a single bounded coefficient is therefore estimated from calibration pairs

$$\{(r_{i,h}, \widehat{r}_{i,h})\}_{i \in \mathcal{D}_{\mathrm{cal}}}. \tag{68}$$

The calibration objective is

$$\mathcal{L}_h(\gamma) = \sum_{i \in \mathcal{D}_{\mathrm{cal}}} \left(r_{i,h} - \gamma \widehat{r}_{i,h}\right)^2. \tag{69}$$

Differentiating gives

$$\frac{\partial \mathcal{L}_h}{\partial \gamma} = -2 \sum_{i \in \mathcal{D}_{\mathrm{cal}}} \widehat{r}_{i,h} \left(r_{i,h} - \gamma \widehat{r}_{i,h}\right), \tag{70}$$

$$0 = \sum_{i \in \mathcal{D}_{\mathrm{cal}}} \widehat{r}_{i,h} r_{i,h} - \gamma \sum_{i \in \mathcal{D}_{\mathrm{cal}}} \widehat{r}_{i,h}^2. \tag{71}$$

The unconstrained solution is

$$\widetilde{\gamma}_h = \frac{\sum_{i \in \mathcal{D}_{\mathrm{cal}}} \widehat{r}_{i,h} r_{i,h}}{\sum_{i \in \mathcal{D}_{\mathrm{cal}}} \widehat{r}_{i,h}^2 + \varepsilon_\gamma}. \tag{72}$$

The final coefficient is

$$\widehat{\gamma}_h = \Pi_{[0,\gamma_{\max}]}\left(\widetilde{\gamma}_h\right), \tag{73}$$

where the Euclidean projection is

$$\Pi_{[0,\gamma_{\max}]}(x) = \min\left\{\gamma_{\max}, \max\{0, x\}\right\}. \tag{74}$$

The lower boundary suppresses a residual correction whose calibration association is non-positive, whereas the upper boundary prevents excessive amplification. Only one scalar is estimated per horizon, thereby limiting calibration variance without requiring nonlinear gate optimization. The FastSketch forecast is

$$\widehat{y}^{\mathrm{FS}}_{t+h|t} = \widehat{y}^{\mathrm{base}}_{t+h|t} + \widehat{\gamma}_h \widehat{r}_{t+h|t}, \qquad h = 1, \ldots, H. \tag{75}$$

The reported experiments use $H = 1$. The multi-horizon notation is retained to show that the architecture extends directly by assigning an independent readout column and shrinkage coefficient to each horizon.

2.5. Adaptive Conformal Uncertainty Estimation

The same uncertainty-estimation layer is applied after either S3-Forecaster or S3-FastSketch. Let $\widehat{y}_{t|t-1}$ denote the final point forecast. The nonconformity score is

$$s_t = \left|y_t - \widehat{y}_{t|t-1}\right|. \tag{76}$$

For $H > 1$, a simultaneous score can be defined as

$$s_t^{(H)} = \max_{h=1,\ldots,H} \left|y_{t+h} - \widehat{y}_{t+h|t}\right|. \tag{77}$$

2.5.1. Dynamic Score Buffer

Let W denote the maximum calibration-buffer length. Immediately before forecasting y_t, the score buffer is

$$\mathcal{S}_t = \{s_j : \max(t-W, t_{\mathrm{cal},1}) \leq j \leq t-1\}, \tag{78}$$

where $t_{\mathrm{cal},1}$ is the first calibration index. Thus,

$$|\mathcal{S}_t| \leq W. \tag{79}$$

Given an effective miscoverage level δ_t, define

$$k_t = \min\{|\mathcal{S}_t|, \lceil(|\mathcal{S}_t|+1)(1-\delta_t)\rceil\}. \tag{80}$$

The finite-buffer empirical quantile is

$$q_t(\delta_t) = s_{(k_t)}(\mathcal{S}_t), \tag{81}$$

where $s_{(k_t)}(\mathcal{S}_t)$ is the k_t-th order statistic. The adaptive conformal interval is

$$\widehat{C}_t = \left[\widehat{y}_{t|t-1} - q_t(\delta_t),\ \widehat{y}_{t|t-1} + q_t(\delta_t)\right]. \tag{82}$$

2.5.2. Sequential Miscoverage Update

After observing y_t, define

$$M_t = \mathbb{I}\left\{y_t \notin \widehat{C}_t\right\}. \tag{83}$$

Let $\delta^\star$ be the target miscoverage, with $\delta^\star = 0.10$ for nominal 90% coverage. The effective miscoverage is updated as

$$\delta_{t+1} = \Pi_{[\delta_{\min},\delta_{\max}]}\left[\delta_t + \eta_{\mathrm{ACI}}\left(\delta^\star - M_t\right)\right], \tag{84}$$

where

$$\Pi_{[\delta_{\min},\delta_{\max}]}(x) = \min\{\delta_{\max}, \max\{\delta_{\min}, x\}\}. \tag{85}$$

When a miss occurs, $M_t = 1$, the update decreases δ_{t+1} and therefore raises the next empirical quantile, producing a wider interval. When coverage is repeatedly achieved, δ_t can increase gradually, reducing excessive width. Finally,

$$\mathcal{S}_{t+1} = \mathrm{Tail}_W\left(\mathcal{S}_t \cup \{s_t\}\right). \tag{86}$$

This construction does not assume a Gaussian residual distribution. Its distribution-free interpretation concerns the absence of a parametric error law under conformal validity conditions. Under serial dependence and temporal drift, ACI targets adaptive long-run miscoverage control; it does not imply exact conditional coverage for every individual forecast or every short test block.

2.6. Model Complexity

For S3-Forecaster, the number of learned readout coefficients is approximately

$$P_{\mathrm{S3}} = H(1 + R + p) + 2, \tag{87}$$

where the final two parameters correspond to the OOB gate. The fixed matrices $\boldsymbol{W}_{\mathrm{in}}$ and $\boldsymbol{W}_{\mathrm{res}}$ are excluded because they are not trained.

For S3-FastSketch,

$$P_{\mathrm{FS}} = HD_{\phi} + H, \tag{88}$$

where the final H coefficients are the calibration shrinkage factors. For a dense reservoir, state propagation requires

$$\mathcal{O}(NR^2) \tag{89}$$

operations, which can be reduced using a sparse recurrent matrix. FastSketch feature construction requires approximately

$$\mathcal{O}(ND_{\phi}). \tag{90}$$

Both methods avoid iterative backpropagation through a nonlinear sequence model. Their learned readouts are obtained through convex regularized objectives, while FastSketch additionally avoids nonlinear gate calibration.

3. Experimental Setup

3.1. Datasets and Natural Small-Data Cohorts

The empirical evaluation comprises four public monthly forecasting collections—M3 Monthly, M4 Monthly, Tourism Monthly, and CIF 2016—and an anonymized Industrial Construction-Materials Demand dataset, hereafter denoted ICMD. The same experimental procedure is applied independently to every dataset.

The public collections cover heterogeneous forecasting domains, including macroeconomic, demographic, tourism-demand, banking, and general competition series. ICMD contains 61,395 transactional records, 47 original variables, 555 product references, and 62 monthly periods between January 2020 and February 2025. For each eligible product reference i, the monthly target is constructed as

$$y_{i,t} = \sum_{j \in \mathcal{T}_{i,t}} \mathrm{SalesValue}_j, \tag{91}$$

where $\mathcal{T}_{i,t}$ denotes the set of valid positive-value transactions associated with product i during month t. In contrast to a single-reference case study, ICMD is evaluated as a multi-series cohort containing all product-level histories that satisfy the predefined eligibility criteria.

The primary benchmark is restricted to naturally short series. A series is eligible when it has monthly frequency, finite observations, a sufficient unmodified history for chronological training and calibration, and fewer than 200 available pre-test observations. Longer histories are excluded rather than artificially truncated. Consequently, the main benchmark evaluates naturally occurring small-data conditions. Controlled reductions of the available history are introduced only in the Data Efficiency experiment described in Section 3.9.

When a collection contains more than 200 eligible series, at most 200 are selected using a deterministic hash-based ordering with seed 42. This selection is performed before model execution and is independent of forecasting performance. All eligible series are retained when the collection contains fewer than 200 candidates. Series identifiers, original lengths, selected development series, and exclusion reasons are stored with the experimental outputs.

3.2. Development, Failure, and Ranking Cohorts

Let $\mathcal{E}_d$ denote the eligible series in dataset d, and let $\mathcal{D}_d \subset \mathcal{E}_d$ denote the series reserved for dataset-level model selection. Development series are excluded from the final benchmark.

Model failures are not silently removed from the experimental record. For each model m, let

$$\mathcal{S}_{m,d} = \{i \in \mathcal{E}_d \setminus \mathcal{D}_d : m \text{ completes the evaluation of series } i\} \tag{92}$$

denote its successful evaluation cohort. The common cohort used for the primary dataset-level ranking is

$$\mathcal{C}_d = (\mathcal{E}_d \setminus \mathcal{D}_d) \cap \bigcap_{m \in \mathcal{M}} \mathcal{S}_{m,d}, \tag{93}$$

where $\mathcal{M}$ is the complete model set. Thus, every model is ranked using exactly the same series. Model-specific failures, exception messages, and failure rates are reported separately, together with the sizes of $\mathcal{E}_d$, $\mathcal{D}_d$, $\mathcal{S}_{m,d}$, and $\mathcal{C}_d$. Dataset-level metrics are not pooled across collections with different scales.

3.3. Reference Models

The proposed S3-Forecaster and S3-FastSketch are compared with reference models spanning complementary forecasting paradigms:

1. **Statistical forecasters:** Naive, Seasonal Naive, AutoRegressive (AR), ARIMA, ETS, and Theta;
2. **Kernel methods:** Kernel Ridge Regression (KRR) and Gaussian Process Regression (GPR);
3. **Lightweight neural forecasters:** EasyTSF-compatible DLinear and NLinear models, a causal one-dimensional CNN, and a unidirectional LSTM;
4. **Time-series foundation models:** Chronos-Bolt-Small, using the `amazon/chronos-bolt-small` checkpoint, and TimesFM-2.5-200M, using `google/timesfm-2.5-200m-pytorch`;
5. **Proposed models:** S3-Forecaster and S3-FastSketch.

DLinear, NLinear, CNN, and LSTM use chronological input windows and a temporal validation suffix contained entirely within the training data. Neural samples are not shuffled, and early stopping is determined without access to calibration or test observations. Chronos and TimesFM remain frozen and are evaluated in zero-shot mode; no target-domain gradient update is applied to either foundation model.

Prophet is excluded a priori because repeated fitting over several hundred series and rolling evaluation steps produces a disproportionate computational cost. Its exclusion is based on operational feasibility rather than observed forecasting accuracy. ETS and Theta preserve the statistical trend and seasonality reference class that Prophet would otherwise partially duplicate.

3.4. Offline Model Selection and Frozen Reuse

Hyperparameter selection and final evaluation are treated as disjoint computational phases. For model m and dataset d, let $\mathcal{G}_{m,d}$ denote the predefined dataset-level search space. The selected configuration is

$$\boldsymbol{\theta}^{\star}_{m,d} = \arg\min_{\boldsymbol{\theta}\in\mathcal{G}_{m,d}} \frac{1}{|\mathcal{D}_d|} \sum_{i\in\mathcal{D}_d} \mathcal{L}_{\mathrm{val}}\left(m, \boldsymbol{\theta}; \mathbf{y}^{(i)}\right). \tag{94}$$

For the S3 models, point-forecast selection uses three chronological validation folds of six observations, 100 optimization trials, and epsilon-stabilized MAPE as the objective. Uncertainty hyperparameters are selected in a subsequent 60-trial study by minimizing

$$\mathcal{J}_{\mathrm{UQ}} = \mathrm{MSIS} + \lambda_{\mathrm{cov}} \left[\max\left\{0,\, 0.90 - \mathrm{ECP}\right\}\right]^2. \tag{95}$$

After selection, $\boldsymbol{\theta}^{\star}_{m,d}$ is serialized and reused without modification in the main benchmark, ablation, Shock/Non-Shock, Data Efficiency, and Multi-prior experiments. No per-series or test-dependent retuning is permitted. The one-time HPO cost is reported separately and is not included in operational forecasting time.

3.5. Chronological Splitting and Rolling Evaluation

For a series $\mathbf{y}^{(i)} = (y_1^{(i)}, \dots, y_{N_i}^{(i)})$, the chronological boundaries are

$$n_{\mathrm{tr}}^{(i)} = \lfloor 0.64 N_i \rfloor, \tag{96}$$

$$n_{\mathrm{dev}}^{(i)} = \lfloor 0.80 N_i \rfloor, \tag{97}$$

$$n_{\mathrm{cal}}^{(i)} = n_{\mathrm{dev}}^{(i)} - n_{\mathrm{tr}}^{(i)}, \tag{98}$$

$$n_{\mathrm{te}}^{(i)} = N_i - n_{\mathrm{dev}}^{(i)}. \tag{99}$$

These boundaries define disjoint training, calibration, and test blocks:

$$\mathcal{I}_{\mathrm{tr}}^{(i)} = \left\{1, \dots, n_{\mathrm{tr}}^{(i)}\right\}, \tag{100}$$

$$\mathcal{I}_{\mathrm{cal}}^{(i)} = \left\{n_{\mathrm{tr}}^{(i)} + 1, \dots, n_{\mathrm{dev}}^{(i)}\right\}, \tag{101}$$

$$\mathcal{I}_{\mathrm{te}}^{(i)} = \left\{n_{\mathrm{dev}}^{(i)} + 1, \dots, N_i\right\}. \tag{102}$$

All models are evaluated under one-step-ahead rolling forecasting with horizon $H = 1$ and rolling step equal to one. For every $t \in \mathcal{I}_{\mathrm{te}}^{(i)}$, the forecast and interval satisfy

$$\widehat{y}_{t|t-1},\, \widehat{L}_{t|t-1},\, \widehat{U}_{t|t-1} \in \mathcal{F}_{t-1}, \tag{103}$$

where $\mathcal{F}_{t-1}$ contains only information available before observing y_t. Once the prediction has been recorded, y_t may be used to update the causal state and conformal buffer for step $t+1$. Dataset-level hyperparameters remain frozen throughout the test block.

Preprocessing statistics are estimated exclusively from the training block. Forecasts and interval bounds are inverse-transformed before evaluation, and all reported errors are computed on the original data scale.

3.6. Common Adaptive Conformal Wrapper

To ensure a homogeneous uncertainty comparison, the same Adaptive Conformal Inference wrapper is applied to every point forecaster, including statistical, kernel, neural, foundation, and S3 models. Native intervals from probabilistic baselines are not used in the primary comparison; they may be reported separately as a supplementary sensitivity analysis.

Let

$$s_t = \left| y_t - \widehat{y}_{t|t-1} \right| \tag{104}$$

denote the absolute nonconformity score, and let q_t be the empirical $1 - \alpha_t$ quantile of the causally available conformity buffer. The prediction interval is

$$\widehat{C}_t = \left[\widehat{y}_{t|t-1} - q_t,\, \widehat{y}_{t|t-1} + q_t \right]. \tag{105}$$

For target miscoverage $\alpha = 0.10$, the adaptive level is updated as

$$\alpha_{t+1} = \Pi_{\mathcal{A}} \left[\alpha_t + \gamma \left(\alpha - \mathbb{I}\{ y_t \notin \widehat{C}_t \} \right) \right], \tag{106}$$

where γ is the frozen adaptation step and $\Pi_{\mathcal{A}}$ projects the update onto the admissible range. This protocol targets a nominal coverage of 90% without allowing future test observations to affect the current interval.

3.7. Component-Wise Ablation Study

Ablation experiments reuse the frozen point and uncertainty configurations. Each ablated variant is evaluated on the same chronological partitions and paired series as its corresponding complete model.

For S3-Forecaster, the evaluated variants remove the residual adapter, reservoir states, autoregressive lags, volatility gate, or adapter selector. A static conformal variant is obtained by setting the ACI adaptation step to zero.

For S3-FastSketch, the variants remove the residual adapter, autoregressive lags, EMA residual features, volatility features, momentum features, causal convolutions, seasonal features, calendar features, the complete multiscale sketch, calibration shrinkage, or the adapter selector. A static conformal variant is also included.

For series i, model family a, and ablation v, the paired loss difference is

$$\Delta\mathcal{L}_{a,v}^{(i)} = \mathcal{L}_{a,v}^{(i)} - \mathcal{L}_{a,\mathrm{full}}^{(i)}. \tag{107}$$

Positive values indicate degradation relative to the complete architecture. Ablation summaries are calculated over the common set of series completed by the full model and all compared variants.

3.8. Shock and Non-Shock Analysis

Series are assigned to Shock or Non-Shock regimes using training observations only. For monthly seasonality $m = 12$, the seasonal increment is

$$d_t^{(i)} = y_t^{(i)} - y_{t-m}^{(i)}, \qquad t \in \mathcal{I}_{\mathrm{tr}}^{(i)}. \tag{108}$$

A robust location and scale are estimated as

$$c_i = \mathrm{median}_t \left(d_t^{(i)} \right), \tag{109}$$

$$\sigma_i = 1.4826 \ \mathrm{median}_t \left| d_t^{(i)} - c_i \right|. \tag{110}$$

When the MAD scale is numerically degenerate, IQR/1.349 is used as a fallback. The robust shock score is

$$z_t^{(i)} = \frac{\left| d_t^{(i)} - c_i \right|}{\max(\sigma_i, \varepsilon_\sigma)}. \tag{111}$$

Observations satisfying $z_t^{(i)} > 3.5$ are flagged, and consecutive flags are collapsed into a single event. A series is assigned to the Shock regime when at least one such event is present in its training block; otherwise, it is assigned to the Non-Shock regime. The label is fixed before test evaluation.

This experiment is therefore a series-stratified robustness analysis rather than a retrospective classification of test observations. For each model, the regime penalty is summarized through

$$R_m^{\mathrm{shock}} = \frac{\widetilde{\mathcal{L}}_{m,\mathrm{shock}}}{\widetilde{\mathcal{L}}_{m,\mathrm{nonshock}} + \varepsilon}, \tag{112}$$

where $\widetilde{\mathcal{L}}$ denotes the median per-series loss within the corresponding regime. Point accuracy, empirical coverage, interval width, and MSIS are reported separately for Shock and Non-Shock series.

3.9. Data Efficiency Analysis

Data efficiency is evaluated at retained-history fractions

$$\mathcal{R} = \{0.25,\ 0.50,\ 0.75,\ 1.00\}\,. \tag{113}$$

For each fraction r, only the most recent $\lceil r N_i^{\mathrm{pre}} \rceil$ observations of the pre-test history are retained, while the test block, random seed, model configuration, and series identifier remain unchanged. The $r = 1$ condition reuses the full-history benchmark.

A common admissible cohort is determined before executing any fraction. A series is retained only when the smallest history fraction satisfies the seasonal and context-length requirements of every compared model:

$$\lceil r_{\min} N_i^{\mathrm{pre}} \rceil \geq \max\{2m,\ c_{\max} + H + 1\}\,, \tag{114}$$

where $c_{\max}$ is the maximum required input context. All reference and proposed models are evaluated in this analysis. Failed model–fraction cells are explicitly reported and are excluded from comparative curves rather than being silently aggregated.

3.10. Multi-Prior Robustness

The dependence of the S3 architectures on the baseline prior is assessed without re-optimizing their residual adapters. Let $p^\star$ denote the prior selected during HPO. The evaluated prior set is

$$\mathcal{P} = \{p^\star,\ \text{causal rolling mean, Seasonal Naive, ETS, Theta, Chronos, TimesFM}\}\,. \tag{115}$$

For every $p \in \mathcal{P}$, only the prior definition and its required fixed parameters are replaced. Reservoir size, residual lags, sketch features, regularization, shrinkage, gate configuration, calibration split, and ACI hyperparameters remain frozen. Prophet is not included because its repeated rolling execution would impose a disproportionately high computational burden.

The sensitivity of architecture a to prior p is measured by

$$\Delta\mathcal{L}_{a,p}^{(i)} = \mathcal{L}_{a,p}^{(i)} - \mathcal{L}_{a,p^\star}^{(i)}\,. \tag{116}$$

The analysis reports the median and interquartile range of these differences, the percentage of improved series, the worst-performing prior, between-prior dispersion, and the number of failed evaluations.

3.11. Evaluation Metrics and Dataset-Level Aggregation

Let $\mathcal{C}_d$ denote the common evaluation cohort for dataset d, and let $\mathcal{T}_i$ be the test set of series i, with $H_i = |\mathcal{T}_i|$. For each $t \in \mathcal{T}_i$, $y_{i,t}$ denotes the observed value, $\widehat{y}_{i,t}$ the one-step-ahead point forecast, and $[\widehat{L}_{i,t}, \widehat{U}_{i,t}]$ the corresponding prediction interval. The forecast error is defined as

$$e_{i,t} = y_{i,t} - \widehat{y}_{i,t}. \tag{117}$$

All metrics are first computed independently for each model–series pair. Dataset-level statistics are subsequently obtained across the common cohort $\mathcal{C}_d$, thereby assigning equal weight to every series rather than pooling observations from series with different test lengths.

3.11.1. Point-Forecast Metrics

The epsilon-stabilized Mean Absolute Percentage Error is

$$\mathrm{MAPE}_{\varepsilon}^{(i)} = \frac{100}{H_i} \sum_{t \in \mathcal{T}_i} \frac{|e_{i,t}|}{\max\left(|y_{i,t}|, \varepsilon\right)}, \qquad \varepsilon = 10^{-5}. \tag{118}$$

The symmetric Mean Absolute Percentage Error is defined as

$$\mathrm{sMAPE}_{\varepsilon}^{(i)} = \frac{100}{H_i} \sum_{t \in \mathcal{T}_i} \frac{2|e_{i,t}|}{\max\left(|y_{i,t}| + |\widehat{y}_{i,t}|, \varepsilon\right)}. \tag{119}$$

To construct a scale-independent absolute error, let $\mathbf{y}_i^{\mathrm{pre}} = (y_{i,1}, \ldots, y_{i,n_i^{\mathrm{pre}}})$ denote the complete pre-test history available for fitting and calibration. The seasonal naive scaling factor is

$$s_i = \frac{1}{n_i^{\mathrm{pre}} - m} \sum_{t=m+1}^{n_i^{\mathrm{pre}}} \left| y_{i,t} - y_{i,t-m} \right|, \tag{120}$$

where $m = 12$ for the monthly datasets. The Mean Absolute Scaled Error is then

$$\mathrm{MASE}^{(i)} = \frac{\dfrac{1}{H_i} \displaystyle\sum_{t \in \mathcal{T}_i} |e_{i,t}|}{s_i}. \tag{121}$$

A value MASE < 1 indicates that the evaluated model produces a lower mean absolute error than the seasonal naive reference associated with the scaling denominator.

The Root Mean Squared Error is computed on the original data scale as

$$\mathrm{RMSE}^{(i)} = \sqrt{\frac{1}{H_i} \sum_{t \in \mathcal{T}_i} e_{i,t}^2}. \tag{122}$$

MAPE and sMAPE are reported as percentages, MASE provides a scale-free comparison against the seasonal naive reference, and RMSE retains sensitivity to large forecast errors. Consequently, these measures are interpreted jointly rather than as interchangeable estimates of predictive performance.

3.11.2. Prediction-Interval Metrics

All uncertainty metrics are evaluated at nominal coverage $1 - \alpha = 0.90$, with $\alpha = 0.10$. The Empirical Coverage Probability for series i is

$$\mathrm{ECP}^{(i)} = \frac{1}{H_i} \sum_{t \in \mathcal{T}_i} \mathbb{I}\left\{ \widehat{L}_{i,t} \le y_{i,t} \le \widehat{U}_{i,t} \right\}. \tag{123}$$

The mean prediction-interval width is

$$\mathrm{MW}^{(i)} = \frac{1}{H_i} \sum_{t \in \mathcal{T}_i} \left(\widehat{U}_{i,t} - \widehat{L}_{i,t} \right). \tag{124}$$

Because coverage alone can be increased by constructing arbitrarily wide intervals, interval calibration and sharpness are evaluated jointly through the interval score. For observation t, the interval score is

$$\begin{aligned} \mathrm{IS}_{\alpha,i,t} = & \left(\widehat{U}_{i,t} - \widehat{L}_{i,t} \right) \\ & + \frac{2}{\alpha} \left(\widehat{L}_{i,t} - y_{i,t} \right) \mathbb{I}\left\{ y_{i,t} < \widehat{L}_{i,t} \right\} \\ & + \frac{2}{\alpha} \left(y_{i,t} - \widehat{U}_{i,t} \right) \mathbb{I}\left\{ y_{i,t} > \widehat{U}_{i,t} \right\}. \end{aligned} \tag{125}$$

The Mean Scaled Interval Score is obtained by normalizing the mean interval score using the same seasonal naive scale employed by MASE:

$$\mathrm{MSIS}^{(i)} = \frac{\frac{1}{H_i} \sum_{t \in \mathcal{T}_i} \mathrm{IS}_{\alpha,i,t}}{s_i}. \tag{126}$$

Lower MSIS values indicate sharper intervals with smaller penalties for observations falling outside the interval. ECP is therefore interpreted relative to the nominal value of 0.90 and jointly with mean width and MSIS; overcoverage is not considered beneficial when it is obtained through excessive interval inflation.

3.11.3. Dataset-Level Aggregation

For any per-series metric $x_{j,i}$ associated with model j and series i, the dataset-level median is

$$\widetilde{x}_{j,d} = Q_{0.50}\left(\{ x_{j,i} : i \in \mathcal{C}_d \} \right), \tag{127}$$

and its interquartile range is

$$\mathrm{IQR}(x_{j,d}) = Q_{0.75}\left(\{ x_{j,i} : i \in \mathcal{C}_d \} \right) - Q_{0.25}\left(\{ x_{j,i} : i \in \mathcal{C}_d \} \right). \tag{128}$$

MAPE, sMAPE, MASE, RMSE, mean interval width, MSIS, trainable parameters, and elapsed time are summarized using the median and IQR. Since coverage is a bounded probability with a direct nominal target, dataset-level ECP is calculated as the unweighted mean of the per-series coverage values:

$$\overline{\mathrm{ECP}}_{j,d} = \frac{1}{|\mathcal{C}_d|} \sum_{i \in \mathcal{C}_d} \mathrm{ECP}_{j,i}. \tag{129}$$

The reported number of series is

$$n_{\mathrm{series},d} = |\mathcal{C}_d|, \tag{130}$$

which is identical across models in the primary ranking because all comparisons use the common successful cohort.

Let $P_{j,i}$ denote the number of target-domain trainable parameters and let $\tau_{j,i}$ denote the elapsed time required for final fitting and rolling prediction. Computational summaries are defined as

$$\widetilde{P}_{j,d} = \mathrm{median}_{i \in \mathcal{C}_d} P_{j,i}, \tag{131}$$

$$\widetilde{\tau}_{j,d} = \mathrm{median}_{i \in \mathcal{C}_d} \tau_{j,i}. \tag{132}$$

Frozen foundation-model weights and fixed random projections are excluded from $P_{j,i}$ because they are not adjusted using the target series. Their total parameter footprint is reported separately. Elapsed time excludes environment installation, checkpoint downloading, and offline hyperparameter optimization.

Finally, the dataset-specific ordinal ranking is obtained by sorting the models in ascending order of median $\mathrm{MAPE}_\varepsilon$:

$$\mathrm{Rank}_{j,d} = \mathrm{rank}_{\uparrow}\left(\widetilde{\mathrm{MAPE}}_{\varepsilon,j,d}\right). \tag{133}$$

The ranking is interpreted together with sMAPE, MASE, RMSE, ECP, mean width, and MSIS, since no individual metric fully characterizes point accuracy, robustness to scale, and uncertainty quality.

Operational runtime includes final fitting and rolling forecasting after loading the frozen configuration. HPO, checkpoint downloading, and environment installation are excluded and reported separately. Trainable-parameter counts include coefficients estimated from the target series but exclude frozen foundation-model weights and fixed random feature matrices. The total parameter footprint of frozen pretrained models is reported separately to avoid interpreting zero target-domain trainable parameters as zero computational complexity.

3.12. Reproducibility Controls

All experiments use fixed random seeds and repository commits. Dataset identifiers, natural series lengths, development exclusions, common cohorts, frozen configurations, model failures, per-series forecasts, and per-series metrics are persisted before aggregation. Neural models are trained without temporal shuffling, and no test observation is used for preprocessing, model selection, early stopping, gate calibration, or conformal initialization. These controls ensure that all reported rankings correspond to a common causal evaluation protocol rather than to model-specific subsets or retrospectively selected histories.

Table 1: Industrial dataset profile.

Characteristic	Value
Transactional records	61,395
Original variables	47
Product references	555
Database coverage	Jan. 2020–Feb. 2025
Monthly periods in database	62
Median active months per product	≈ 6
Evaluated series coverage	Jan. 2020–Dec. 2023
Evaluated monthly observations	48
Final test observations	10
Forecast target	Monthly aggregated sales value

4. Results and Discussion

4.1. Engineering Case Characterization and Non-Stationarity

The primary engineering application is an anonymized *Industrial Construction-Materials Demand* dataset, hereafter ICMD. The source database contains heterogeneous transactional records rather than a pre-assembled forecasting benchmark. It comprises 61,395 records, 47 variables, 555 product references, and 62 monthly periods between January 2020 and February 2025. Product-level histories are markedly sparse: the median product is observed in only approximately six monthly periods. This catalog-level sparsity is operationally relevant because it prevents the routine use of high-capacity models for most references and motivates a forecasting framework whose trainable complexity remains controlled when $N < 200$.

The target was constructed by chronologically aggregating the positive sales value of the selected construction-material reference at monthly frequency. The evaluated contiguous series contains 48 observations from January 2020 to December 2023, of which the last 10 months constitute the final test block. The company identity and product identifier are omitted for confidentiality, while the aggregation rule, temporal split, target definition, and model code remain fully specified. Table 1 summarizes the resulting data audit.

Figure 1 shows a changing level and substantial short-range variation throughout the observed window. The dominant empirical property is therefore non-stationarity in location and local scale, rather than the presence of a threshold-defined extreme event in the final test period. This distinction is central to the present interpretation: the experiments evaluate adaptation under short, non-stationary histories, but do not claim test-time evidence for abrupt structural discontinuities.

Because the anonymized release does not expose the raw product-level industrial series, the diagnostic values in Table 2 are reported as audit fields rather than inferred from forecast outputs. They should be computed directly from the confidential monthly target before final submission; until then, the interpretation below is restricted to the visible series profile and benchmark outcomes.

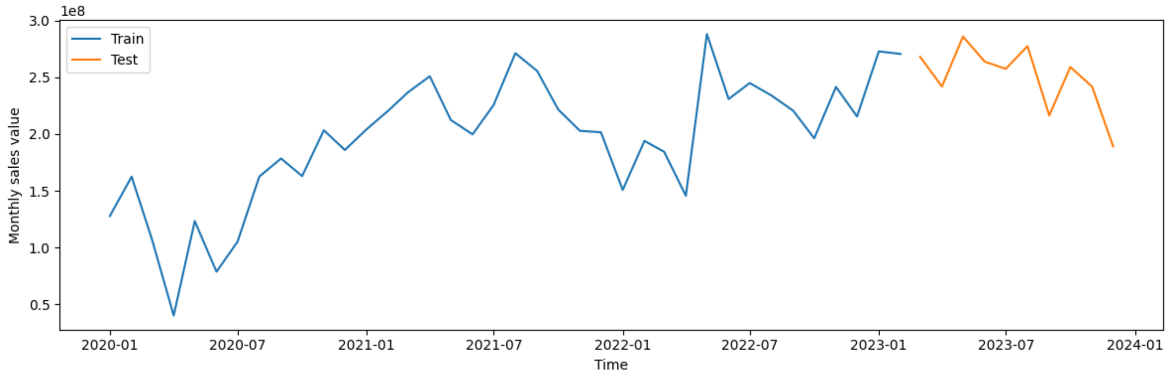

Figure 1: Industrial demand series and temporal split.

Table 2: Industrial series diagnostics.

Diagnostic	Raw scale	Log scale
ADF p-value	*to be audited*	*to be audited*
KPSS p-value	*to be audited*	*to be audited*
Coefficient of variation	*to be audited*	*to be audited*
Lag-1 autocorrelation	*to be audited*	*to be audited*
Lag-12 autocorrelation	*to be audited*	*to be audited*
Seasonal strength	*to be audited*	*to be audited*
Trend strength	*to be audited*	*to be audited*

4.2. Overall Benchmarking and Metric-Specific Behaviour

5. Conclusions

CRediT authorship contribution statement

Alejandro Patino Bedoya: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The M3, M4, Tourism, and CIF 2016 benchmark datasets used in this study are publicly available from the original sources cited in the manuscript. The source code, selected public-series identifiers, truncation lengths, random seeds, serialized hyperparameter configurations, and scripts required to reproduce the public benchmark, ablation, data-efficiency, and multi-prior experiments are openly available at `https://github.com/AlejoPatigno/S3Forecaster-S3FastSketch`.

The Industrial Construction-Materials Demand dataset cannot be made publicly available because the underlying transactional records contain confidential commercial information and are subject to data-use restrictions imposed by the data owner. The manuscript reports the target definition, monthly aggregation procedure, chronological partitions, preprocessing protocol, model configurations, and evaluation procedure required to reproduce the methodological workflow. Access to the proprietary records cannot be guaranteed and would require explicit authorization from the data owner.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI’s ChatGPT to support literature organization, language refinement, and structural editing. After using this tool, the author reviewed, verified, and edited the content as needed and takes full responsibility for the content of the published article.

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